# DS 3010

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# Module 3: Model Selection

## Part 2: Best Subset Selection + Cross Validation

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In order to select the best model with respect to test error, we need to estimate this test error.

1. We can indirectly estimate test error by making an adjustment to the training error to account for the bias due to overfitting. E.g. AIC, BIC… (Note, so far, in best subset selection, each model is fit using the entire dataset)

2. We can directly estimate the test error, using either a validation set approach or a cross-validation approach.
   - **Validation set approach**
   - **Cross-Validation**

# Best Subset Selection + Validation Set Approach

**Step 0: Split dataset into training (50%) and test (50%) dataset.**

**Algorithm:**

**Step 1**

k = 0: no predictors , call the model $M_0$

**Step 2**

k =1: $X_1, X_2, X_3, \ldots. X_p$ find best model $M_1$

k = 2, $(X_1, X_2), (X_1, X_3), (X_2, X_3), \ldots$ find best model and $M_2$

k = 3, $(X_1, X_2, X_3)\ldots\ldots.$ find best model and $M_3$

.....

k = p, $(X_1, X_2, X_3, \ldots. X_p)$ find best model $M_k$

**Training MSE**

**Step 2: best subset model is the one with lowest training RSS.**

- Fit the model with **training data**
- Calculate **training MSE.**
- Best model **$M_i$ : model** with lowest **training MSE.**

**Step 3: Evaluate on validation set, final best model is the model that minimizes validation MSE.**

# Best Subset Selection + Validation Set Approach

Drawback of Validation Set Approach:

- Don’t get to see all the data.
- Randomness

## In-class activity:

# Best Subset Selection + K-fold Cross Validation

# Best Subset Selection + K-fold Cross Validation

Divide the dataset into k folds

**Outer loop i:**

- Use fold **i** for test, and the rest folds for training
- Perform Best Subset Selection on Training Data and obtained 19 best subset models
- **Inner loop j:** for all **19** best subset models, calculate test_MSE[i, j]

| Model size | 1 | 2 | .. | 19 |
|---|---|---|---|---|
| Test MSE (test: fold 1) | 45 | 56 | 49 | 89 |
| Test MSE (test: fold 2) | 45 | 56 | 49 | 89 |
| …… | … | … | … | … |
| **Average Test MSE** | 45 | 56 | 49 | 89 |

**Best model size is the model with lowest Average Test MSE**